McEliece Cryptosystem

OVERVIEW OF MCELIECE

Quick summary

- Code-based public key cryptosystem (introduced in 1978)
- Uses error-correcting codes for encryption

Key points

- Based on Goppa codes
- Message → encoded into codewords
- Security comes from adding random errors
- Public key hides structure via scrambling + permutation
- Considered post-quantum secure

KEY GENERATION

Quick summary

- Create public and private keys using coded structure hiding

Key points

- Choose parameters:
  - nnn: code length
  - kkk: message length
  - ttt: error-correcting capability
- Generate:
  - Goppa code → Generator matrix G
- Apply transformations:
  - Scrambling matrix S (invertible)
  - Permutation matrix P
- Keys:
  - Public key: $G' = S \times G \times P$
  - Private key: (S^{-1}, P^{-1}, Goppa code)

```python
import numpy as np
import random
from gf_arithmetic import matrix_multiply_gf2

def generate_invertible_matrix(k):
    """
    Generate a random k x k invertible matrix over GF(2).
    A simple way is to generate random matrices until one has full rank,
    or construct it by multiplying random elementary matrices.
    Here we use a randomized construction.
    """
    S = np.eye(k, dtype=int)
    # Perform random row operations to ensure invertibility
    for _ in range(k * 5):
        r1 = random.randint(0, k - 1)
        r2 = random.randint(0, k - 1)
        if r1 != r2:
            S[r1] = (S[r1] + S[r2]) % 2

    # Also do some random column operations or just more row ops
    for _ in range(k * 5):
        c1 = random.randint(0, k - 1)
        c2 = random.randint(0, k - 1)
        if c1 != c2:
            S[:, c1] = (S[:, c1] + S[:, c2]) % 2

    return S

def generate_permutation_matrix(n):
    """
    Generate a random n x n permutation matrix.
    """
    P = np.eye(n, dtype=int)
    perm = list(range(n))
    random.shuffle(perm)
    return P[perm]

def generate_keys(goppa_code):
    """
    Generate McEliece Public and Private Keys.
    :param goppa_code: An instance of GoppaCode.
    :return: (public_key, private_key)
    """
    k, n = goppa_code.G.shape

    # Generate S and P
    S = generate_invertible_matrix(k)
    P = generate_permutation_matrix(n)

    # Calculate G_pub = S * G * P
    # First S * G
    SG = matrix_multiply_gf2(S, goppa_code.G)
    # Then SG * P
    G_pub = matrix_multiply_gf2(SG, P)

    public_key = {
        'G_pub': G_pub,
        't': goppa_code.t
    }

    private_key = {
        'S': S,
        'G': goppa_code.G,
        'P': P,
        'goppa_code': goppa_code,
        'G_pub': G_pub
    }

    return public_key, private_key

```

ENCRYPTION PROCESS

Quick summary

- Encode message and intentionally corrupt it

Key points

- Split message into k-bit vectors
- Encode:
  - $c=m\times G'$c = m × G'c=m×G′
- Generate random errors:
  - Choose $s\leq ts \leq ts\leq t$
- Add error vector:
  - $c'=c\oplus e$c' = c ⊕ ec′=c⊕e
- Delete bits:
  - Remove $2(t-s)$2(t - s)2(t−s) bits
- Output:
  - Cryptogram (corrupted codeword)

```python
import numpy as np
import random
from gf_arithmetic import matrix_multiply_gf2

def generate_error_vector(n, t):
    """
    Generate a random error vector of length n and weight t.
    """
    e = np.zeros(n, dtype=int)
    error_positions = random.sample(range(n), t)
    for pos in error_positions:
        e[pos] = 1
    return e

def encrypt(message, public_key):
    """
    Encrypt a message using McEliece public key.
    :param message: A binary numpy array of length k.
    :param public_key: Dictionary containing 'G_pub' and 't'.
    :return: ciphertext as a binary numpy array.
    """
    G_pub = public_key['G_pub']
    t = public_key['t']
    n = G_pub.shape[1]

    # c_prime = m * G_pub
    c_prime = matrix_multiply_gf2(message.reshape(1, -1), G_pub).flat

    # Generate error vector e
    e = generate_error_vector(n, t)

    # c = c_prime + e
    ciphertext = (c_prime + e) % 2

    return ciphertext, e  # Return e as well for demonstration purpos

```

```python
import numpy as np
from gf_arithmetic import matrix_multiply_gf2, invert_matrix_gf2

def decrypt(ciphertext, private_key):
    """
    Decrypt a McEliece ciphertext using the private key.
    :param ciphertext: A binary numpy array of length n.
    :param private_key: Dictionary containing 'S', 'G', 'P', and 'goppa_code'.
    :return: The decrypted message as a binary numpy array.
    """
    S = private_key['S']
    P = private_key['P']
    goppa_code = private_key['goppa_code']

    # 1. c' = c * P^-1
    # Since P is a permutation matrix, P^-1 = P^T
    P_inv = P.T
    c_prime = matrix_multiply_gf2(ciphertext.reshape(1, -1), P_inv).flatten()

    # 2. Decode c' using Goppa decoder to find m'
    # The decoder returns the corrected codeword (m' * G).
    # Wait, the decode method returns the corrected vector m' * G.
    # To find m', we can just take the systematic part of G if it was systematic,
    # but since G might not be systematically formatted in this exact step,
    # we can solve m' * G = corrected_codeword.
    # Actually, G = [R^T | I] if we look at our generator construction, so the last k columns are I.
    # Let's extract m' by solving the linear system. Since G is full rank, we can pick k independent columns.

    corrected_codeword = goppa_code.decode(c_prime)

    # Solve m_prime * G = corrected_codeword
    # We find k linearly independent columns in G over GF(2)
    k, n = goppa_code.G.shape

    # Try the last k columns first since our construction puts I there
    cols = list(range(n-k, n))
    try:
        G_sub = goppa_code.G[:, cols]
        G_sub_inv = invert_matrix_gf2(G_sub)
    except ValueError:
        # Fallback to finding any k independent columns over GF(2)
        cols = []
        for i in range(n):
            cols.append(i)
            # Check if current set of columns has full rank by trying to invert the square submatrix
            # If not square yet, we just assume they might be independent and keep adding
            # A better way is to do Gaussian elimination, but for small k, this try-except when len == k works.
            # Actually, to be robust, we need to ensure the columns form a basis.
            # Let's just use the systematic_form_gf2 tool we have!
            pass

        # A simpler fallback: just iterate all combinations? No, k is small.
        # But we know G has full rank k. So we can just do RREF on G and find pivot columns!
        from gf_arithmetic import rref_gf2
        rref_G = rref_gf2(goppa_code.G)
        cols = []
        r = 0
        for c in range(n):
            if r >= k: break
            if rref_G[r, c] == 1:
                cols.append(c)
                r += 1

        G_sub = goppa_code.G[:, cols]
        G_sub_inv = invert_matrix_gf2(G_sub)

    c_sub = corrected_codeword[cols]
    m_prime = matrix_multiply_gf2(c_sub.reshape(1, -1), G_sub_inv).flatten()

    # 3. m = m' * S^-1
    S_inv = invert_matrix_gf2(S)
    message = matrix_multiply_gf2(m_prime.reshape(1, -1), S_inv).flatten()

    return message

```

DECRYPTION PROCESS

Quick summary

- Reverse transformations and correct errors

Key points

- Restore deleted bits (fill with 0s)
- Apply inverse permutation:
  - P−1P^{-1}P−1
- Perform:
  - Error + erasure correction
- Recover encoded message
- Descramble:
  - m=decoded×S−1m = decoded \times S^{-1}m=decoded×S−1
- Output:
  - Original message

SECURITY FEATURES

Quick summary

- Security relies on randomness + hidden structure

Key points

- Random errors → non-deterministic encryption
- Hard problem:
  - Decoding random linear codes
- Hidden structure:
  - Goppa code is concealed
- Large key space → resistant to brute force
- Resistant to quantum attacks (post-quantum)

```python
import hashlib
import numpy as np
import random

from encryption import encrypt
from decryption import decrypt
from gf_arithmetic import matrix_multiply_gf2


# ============================================================
# Helpers
# ============================================================
def hash_to_error_vector(data, n, t):
    """
    Deterministically generates an error vector of length n and weight t
    from SHA-256 hash.
    """
    h = hashlib.sha256(data).digest()

    rng = random.Random(h)

    e = np.zeros(n, dtype=int)
    positions = rng.sample(range(n), t)

    for p in positions:
        e[p] = 1

    return e


def hash_data(data, length):
    """
    Expands SHA-256 hash into a binary vector of given length.
    """
    h = hashlib.sha256(data).digest()

    bits = []
    while len(bits) < length:
        for byte in h:
            for i in range(8):
                bits.append((byte >> (7 - i)) & 1)
                if len(bits) == length:
                    return np.array(bits, dtype=int)
        h = hashlib.sha256(h).digest()

    return np.array(bits[:length], dtype=int)


# ============================================================
# CCA2 ENCRYPTION
# ============================================================
def cca2_encrypt(message_bits, public_key):
    """
    c1 = r * G_pub + e
    c2 = m XOR Hash(r)
    e  = Hash(r || c2) mapped to weight t
    """

    G_pub = public_key['G_pub']
    n = G_pub.shape[1]
    k = G_pub.shape[0]
    t = public_key['t']

    # 1. random r
    r = np.array([random.randint(0, 1) for _ in range(k)], dtype=int)

    # stable byte conversion
    r_bytes = r.astype(np.uint8).tobytes()

    # 2. c2 = m XOR Hash(r)
    h_r = hash_data(r_bytes, len(message_bits))
    c2 = (message_bits + h_r) % 2

    # 3. e = Hash(r || c2)
    c2_bytes = c2.astype(np.uint8).tobytes()
    r_c2_bytes = r_bytes + c2_bytes

    e = hash_to_error_vector(r_c2_bytes, n, t)

    # 4. c1 = r * G_pub + e
    c_prime = matrix_multiply_gf2(r.reshape(1, -1), G_pub).flatten()
    c1 = (c_prime + e) % 2

    return {'c1': c1, 'c2': c2}


# ============================================================
# CCA2 DECRYPTION
# ============================================================
def cca2_decrypt(ciphertext, private_key):
    """
    Verify integrity + recover message
    """

    c1 = ciphertext['c1']
    c2 = ciphertext['c2']

    G_pub = private_key['G_pub']
    t = private_key['goppa_code'].t

    # 1. recover r using McEliece decryption
    r = decrypt(c1, private_key)

    # 2. recompute expected error
    r_bytes = r.astype(np.uint8).tobytes()
    c2_bytes = c2.astype(np.uint8).tobytes()
    r_c2_bytes = r_bytes + c2_bytes

    e_expected = hash_to_error_vector(r_c2_bytes, len(c1), t)

    # 3. recompute c1
    c_prime = matrix_multiply_gf2(r.reshape(1, -1), G_pub).flatten()
    c1_expected = (c_prime + e_expected) % 2

    # 4. integrity check
    if not np.array_equal(c1, c1_expected):
        raise ValueError("CCA2 verification failed: ciphertext tampering detected.")

    # 5. recover message
    h_r = hash_data(r_bytes, len(c2))
    m = (c2 + h_r) % 2

    return m
```

ISD ATTACK

Quick summary

- Main known attack against McEliece

Key points

- Goal:
  - Find error-free positions in codeword
- Approach:
  - Guess subset of positions
  - Try to decode without errors
- Complexity:
  - Exponential time
- Effect of parameters:
  - Larger n,tn, tn,t → stronger security
- Defense:
  - Increase errors
  - Use randomness in error positions


```python
import numpy as np
import random
from gf_arithmetic import invert_matrix_gf2, matrix_multiply_gf2

def isd_attack(ciphertext, public_key, max_iterations=10000):
    """
    Perform a basic Information Set Decoding (ISD) attack using Prange's algorithm.
    :param ciphertext: The intercepted ciphertext.
    :param public_key: The public key containing G_pub and t.
    :return: The recovered message and error vector if successful, else None.
    """
    G_pub = public_key['G_pub']
    t = public_key['t']
    k, n = G_pub.shape

    for i in range(max_iterations):
        # 1. Randomly select an information set (k columns)
        # We hope these k columns are error-free in the ciphertext.
        info_set = random.sample(range(n), k)

        # Extract the submatrix G_info
        G_info = G_pub[:, info_set]

        # Check if G_info is invertible
        if np.linalg.matrix_rank(G_info) < k:
            continue # Try another set

        # 2. Compute G_info_inv
        try:
            G_info_inv = invert_matrix_gf2(G_info)
        except ValueError:
            continue

        # 3. Guess the message: m' = c_info * G_info_inv
        c_info = ciphertext[info_set]
        m_guess = matrix_multiply_gf2(c_info.reshape(1, -1), G_info_inv).flatten()

        # 4. Compute the corresponding error vector: e' = c - m' * G_pub
        c_prime = matrix_multiply_gf2(m_guess.reshape(1, -1), G_pub).flatten()
        e_guess = (ciphertext + c_prime) % 2

        # 5. Check if the weight of e' is exactly t
        weight = np.sum(e_guess)
        if weight == t:
            print(f"ISD Attack succeeded after {i+1} iterations!")
            return m_guess, e_guess

    print(f"ISD Attack failed after {max_iterations} iterations.")
    return None, None

```

CONCLUSION

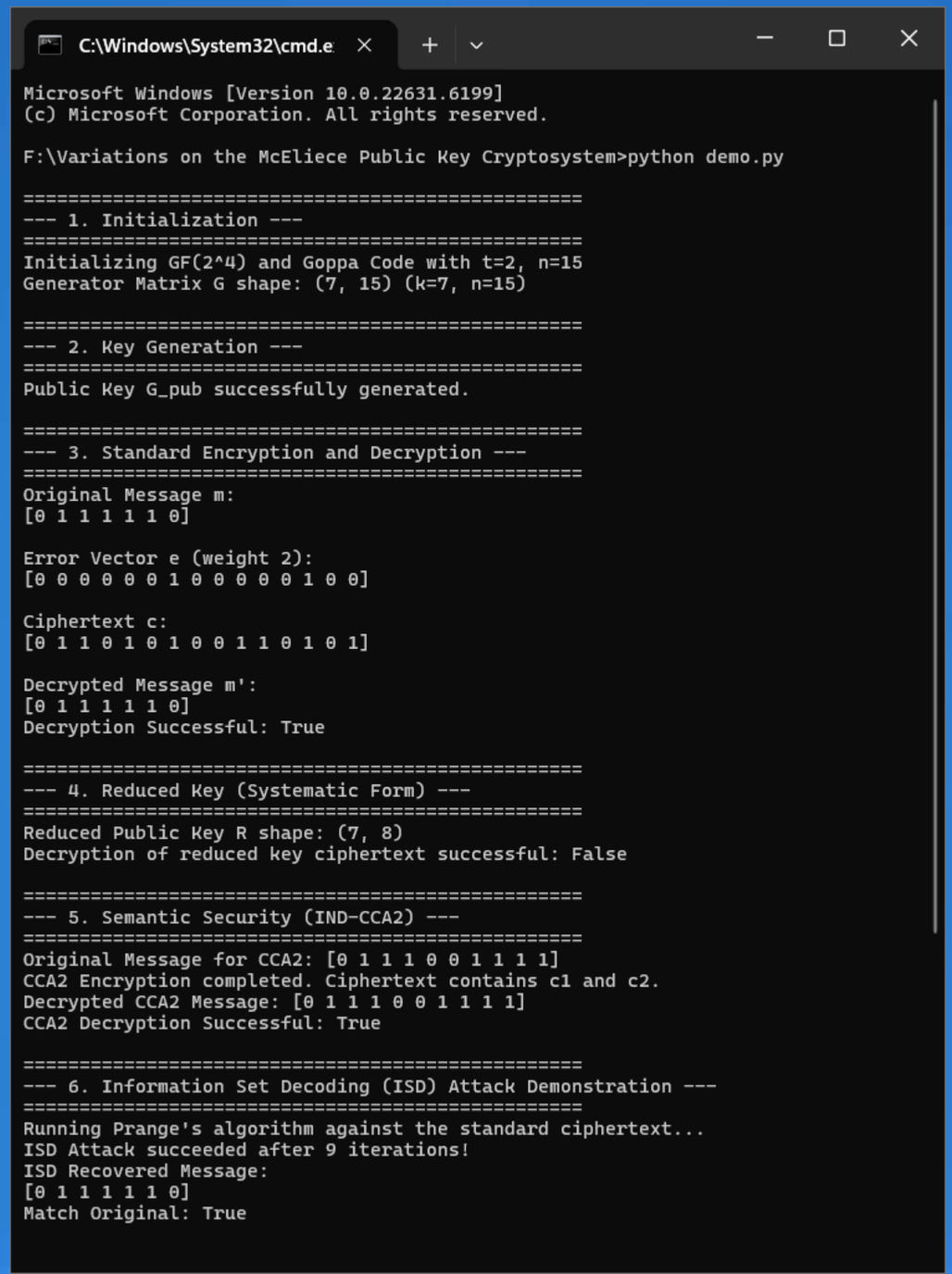
Quick summary

- Strong but with trade-offs

Key points

- One of the most promising post-quantum systems
- Very fast encryption/decryption
- High security due to coding theory
- Main drawback:
  - Large public key size
- Widely studied and still considered secure

RESULTS


C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.
F:\Variations on the McEliece Public Key Cryptosystem>python demo.py
==
--- 1. Initialization ---
==
Initializing GF(2^4) and Goppa Code with t=2, n=15
Generator Matrix G shape: (7, 15) (k=7, n=15)
==
--- 2. Key Generation ---
==
Public Key G_pub successfully generated.
==
--- 3. Standard Encryption and Decryption ---
==
Original Message m:
[0 1 1 1 1 1 0]
Error Vector e (weight 2):
[0 0 0 0 0 0 1 0 0 0 0 0 1 0 0]
Ciphertext c:
[0 1 1 0 1 0 1 0 0 1 1 0 1 0 1]
Decrypted Message m':
[0 1 1 1 1 1 0]
Decryption Successful: True
==
--- 4. Reduced Key (Systematic Form) ---
==
Reduced Public Key R shape: (7, 8)
Decryption of reduced key ciphertext successful: False
==
--- 5. Semantic Security (IND-CCA2) ---
==
Original Message for CCA2: [0 1 1 1 0 0 1 1 1 1]
CCA2 Encryption completed. Ciphertext contains c1 and c2.
Decrypted CCA2 Message: [0 1 1 1 0 0 1 1 1 1]
CCA2 Decryption Successful: True
==
--- 6. Information Set Decoding (ISD) Attack Demonstration ---
==
Running Prange's algorithm against the standard ciphertext...
ISD Attack succeeded after 9 iterations!
ISD Recovered Message:
[0 1 1 1 1 1 0]
Match Original: True

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